

Mahi Aminu Aliyu

Bauchi, Nigeria

mahigital@gmail.com | mahigital.me | LinkedIn

RESEARCH INTERESTS

My research focuses on how machine learning and deep learning models learn meaningful representations, with a particular emphasis on causal representation learning, inductive biases, and reinforcement learning.

EDUCATION

M.Sc., Computer Science, Abubakar Tafawa Balewa University, Nigeria 2023 – Present

Thesis: Location-agnostic Water Quality Model

Advisors: Prof A.Y. Gital & Dr B.I. Ya'u

B.Sc., Computer Science, Al-Qalam University Katsina, Nigeria 2018 – 2023

Thesis: Water Quality Prediction Model

Advisor: Dr Mansir Abubakar

EXPERIENCE

Independent Researcher 2024 – Present

ML Collective (Remote)

Research focus: Causal Representation Learning, ML Theory and Optimization, Multimodal AI, Multilingual datasets.

Data Engineer & Analyst 2023 – Present

Horizon IT Services (Part-time)

- Curated a low-resource Hausa dataset by extracting and cleaning 500+ text entries from Al-Barka radio station's news archives
- Applied advanced data cleaning techniques to transform 2,000+ raw Hausa news entries into a structured CSV dataset
- Collaborated on dataset enhancement by introducing categorization and optional labeling, improving data usability by 30%

PUBLICATIONS & MANUSCRIPTS

[3] **Towards Robust Generalization in African AI: Causal Inference and Domain Shift Mitigation** [Manuscript]

Mahi Aminu Aliyu, Chisom Chibuike, Fatimo Adebajo, Omokolade Awosanya, Samuel Oyeneye

[2] **AfriCaption: Establishing a New Paradigm for Image Captioning in African Languages**, openreview.net [Manuscript]

Mardiyyah Oduwale, Prince Mireku, Fatimo Adebajo, Oluwatosin Olajide, *Mahi Aminu Aliyu*, Jekaterina Novikova

[1] **Water Quality Prediction using USGS Data: A Machine Learning Approach for Automated Assessment Systems** [JISTM '25]

Mahi Aliyu, Mansir Abubakar, Armaya'u Umar Zango, Alwatben Batoul Rashed, Rukayya Musa Ismail

CONFERENCES & SUMMER SCHOOLS

Deep Learning Indaba 2025, Rwanda – Attendee 2025

Selected to attend African largest AI conference with travel grant acquired from ML Collective.

Machine Learning Summer School - Senegal (MLSS) – Participant 2025

Selected to attend advanced ML training led by top global researchers

Deep Learning IndabaX, Nigeria - Attendee 2024

Participated in African-led ML training, discussions, and community-building events

SELECTED PROJECTS

Cross-Lingual Aspect-Based Sentiment Analysis in African Languages

- Translated SemEval-2014 to **17 African languages** (afri-SemEval-2014) with aspect alignment heuristics.
- **Modified** and **fine-tuned** pretrained transformer architectures (e.g., XLM-R, AfroXLM-R) for **causal reasoning** for aspect-based sentiment analysis tasks.

- Causal models showed superior out-of-distribution (OOD) performance with fewer training steps.

SageBot

- Built AI tool that summarizes technical docs for developers, cutting review time by 15%.
- Leveraged LLMs (ChatGPT, LLaMA-3), Hugging Face, and Langchain for high summarization accuracy.
- Improved developer workflow efficiency **by enabling 25%** faster comprehension of complex documents.

Dono Health

- Led development of inclusive insurance MVP with Health Center, Vendor, and Insured modules.
- Designed serverless AWS backend with Lambda, API Gateway, and CDK.
- Integrated Flutterwave for payments and Africa's Talking USSD API for voucher-based funding.

TALKS & PRESENTATIONS

Deep Learning Inbada '25	2025
Poster presentation: "Towards Robust Generalization in African AI: Causal Inference and Domain Shift Mitigation"	
MLSS-Senegal '25	2025
Poster presentation: "Towards Robust Generalization in African AI: Causal Inference and Domain Shift Mitigation"	
ML Collective #25 Research Jam	2025
Oral Presentation: "Pretrained Models for Water-based Applications"	

COMMUNITY SERVICE

City Lead , Data Science Nigeria	2024 – Present
- Organize AI workshops, hands-on sessions, and reading groups for 150+ community members - Promote AI adoption and capacity building across underserved sectors	
DeepLearning.ai Ambassador	2023 – Present
- Facilitated knowledge sharing and engaged 100+ peers as an active community ambassador - Hosted "Growing an AI Community" event in Bauchi with 50+ participants - Organized and participated in 5 cross-community events to strengthen the regional AI ecosystem	
Lead , Google Developer Student Clubs	2021 – 2022
- Led GDSC community, organizing 10+ workshops and technical sessions impacting 350+ students - Applied Google technologies to address UN SDGs, delivering solutions to 5 local communities	

TECHNICAL SKILLS

Programming Languages: Python, C++, R
ML/AI Frameworks: PyTorch, TensorFlow, Hugging Face, Scikit-learn
Web Frameworks & Tools: Django, Flask, Git/GitHub
Cloud Platforms: Microsoft Azure, AWS & GCP
Operating Systems: Linux
Data Analysis: Pandas, NumPy, Matplotlib, Seaborn

RELEVANT COURSES

- Machine Learning Specialization, Coursera – DeepLearning.ai & Stanford	2024
- Statistical Inference, Coursera – Johns Hopkins University	2023
- Data Science: Foundations using R Specialization, Coursera – Johns Hopkins University	2022